

for sustainable growth. Here is our suggestion: When repurchases are important, it is usually simpler to value the firm's total free cash flow, as we did for the concatenation division, rather than just per share dividends. By free cash flow, we mean the cash the firm has available to distribute, either in the form of dividends or of stock repurchases, after paying for all investments necessary for its growth.

In a sense, imagine that you own all the shares in the firm. Free cash flow is what you could receive as the single shareholder in the firm; importantly, it will not depend on the breakdown between dividends and repurchases. After all, as the sole shareholder, you won't care whether you get free cash flow by receiving dividends or by selling some of your shares back to the firm. If you value the whole firm by applying the dividend discount model to free cash flow, then you can find the share price by dividing the value of the firm by the current number of shares outstanding.

7.6 There Are No Free Lunches on Wall Street

Corporations rely on both comparables and the dividend discount model to value entire businesses. Investors in the stock market regularly use comparables in their quest for undervalued stocks. For example, an investor might judge that Kappa Pharmaceuticals is undervalued because it is selling at a lower P/E than Omicron and Upsilon despite the potential profits from its new carboglutin drug. Major investment management companies also use dividend discount models to estimate the value of common stocks and to compare those estimates with stock market prices. In each case, the investor is betting that once others have realized that a stock is undervalued, its price will shoot up and earn her juicy profits.

We will look shortly at how challenging it is for investors to find undervalued stocks. But before we do so, we should explain why a corporate financial manager, who is more concerned with issuing or purchasing securities, should care whether investors can "beat the market." The reason is that, if even professional investors cannot consistently find undervalued stocks, then financial managers should generally assume that market prices provide a good estimate of intrinsic value. A similar message applies to the financial manager when setting the firm's exchange rate policy or deciding on an issue of debt. If it is impossible to identify when a currency or a bond is wrongly valued, then the financial manager should trust market prices rather than engage in risky attempts to "beat the market."

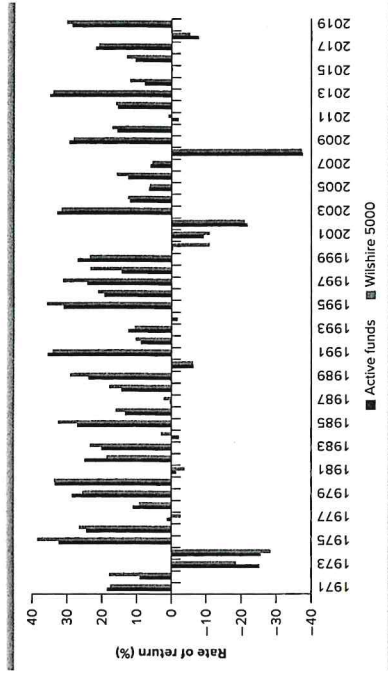
Let's start with a quick look at the performance of amateur investors. Brad Barber and Terrance Odean examined the common stock investments of more than 66,000 households during the period 1991 to 1996.¹² After controlling for differences in risk, they found that the average household underperformed the market by about 1.8% a year before trading costs and by 3.7% after costs.

It may not be surprising that the amateurs don't beat the market, but what about the pros? Presumably, if the amateurs lose before trading costs, the professional investors must gain. Figure 7.6 provides some evidence on the matter. It shows the average performance of equity mutual funds over three decades. You can see that in some years, these mutual funds did beat the market, but more often than not (in fact, in 30 of the 49 years since 1970), it was the other way around. On average, these funds underperformed the market by about 1% a year. It looks as if any gains that the professionals made at the expense of the amateurs were more than eaten up by expenses.

Of course, it would be surprising if some managers were not smarter than others and able to earn superior returns. Unfortunately, it seems difficult to spot the smart ones. For example, a top-quartile fund in one year has no more than an average chance of being in the top quartile the following year. It seems that the top-performing managers in one period have about an average chance of falling on their faces in the next period.

¹² B.M. Barber and T. Odean, "Trading Is Hazardous to Your Wealth: The Stock Market Investment Performance of Individual Investors," *Journal of Finance* 55 (2000), pp. 773–806.

FIGURE 7.6 Annual returns on the Wilshire 5000 Market Index and equity mutual funds, 1971–2019. The market index provided a higher return than the average mutual fund in 30 of the 49 years.



As this kind of discouraging evidence has accumulated, many investors have given up the search for superior investment returns. Instead, they simply buy and hold index mutual funds or exchange-traded funds (ETFs) that track the entire stock market. We discussed index funds and ETFs in Chapter 2. Recall that they provide maximum diversification, with very low management fees. Why pay higher fees to managers who attempt to "beat the market" but can't do so consistently? About half of the equity managed by investment companies (essentially, mutual and exchange-traded funds) is now indexed.

Random Walks and Efficient Markets

Why is it so difficult to earn superior returns? Suppose that you manage a portfolio of common stocks. You try to gauge the business prospects of each firm by studying the financial and trade press, the company's financial accounts, the president's annual statements, and other items of news. You attend company presentations and speak to financial analysts who specialize in a particular industry. Your aim is to uncover undervalued stocks that will offer a superior expected return. But what happens if there are many other talented and industrious analysts trying to find undervalued stocks? If you uncover a stock that appears to be a bargain, it stands to reason that others will also do so, and there will be a wave of buying that pushes up its price. This buying pressure will eliminate the original bargain opportunity.

Suppose some new information becomes available—perhaps Kappa announces that it has obtained FDA approval for its new drug. You rush to buy. The chances are that you will be too late. New information is typically incorporated into stock prices in seconds or minutes. For example, every day the cable television news provider CNBC features a Morning and Midday Call that summarizes the opinions of security analysts and provides information about individual stocks. A study of 322 stocks that were discussed on these calls in 2000 found that positive reports resulted in a significant price impact beginning seconds after the stock is first mentioned and lasting approximately one minute. Investors could have earned a small profit after expenses only if they were able to buy within about 15 seconds of the report.¹³

The price that emerges from this competition is likely to reflect the information and views of all the market participants. In this case, it is not enough to have better information than any one of your competitors; to earn superior returns, you need to have better information than all of them combined. This is a tall order.

¹³ The price response to negative reports was more gradual, lasting 15 minutes, perhaps due to the higher costs of short selling. See J. A. Busse and T. C. Green, "Market Efficiency in Real Time," *Journal of Financial Economics* 65 (2002), pp. 415–437.

FIGURE 7.7a Each dot shows the returns on the New York Composite Index on two successive weeks over a 40-year period. The circled dot shows a weekly return of +3.1%, followed by +5.2% in the next week. The scatter diagram shows no significant relationship between returns on successive weeks.

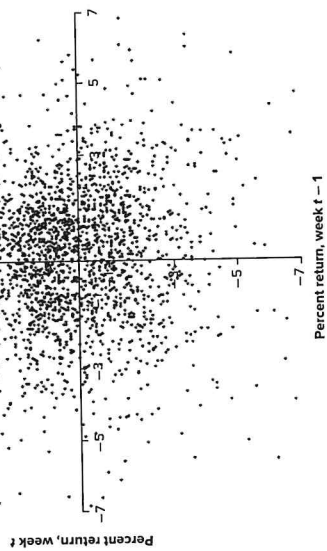
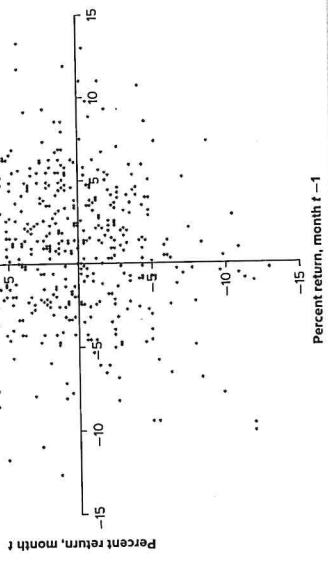


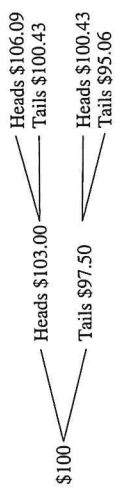
FIGURE 7.7b This scatter diagram shows that there is also no relationship between market returns in successive months.



Professional investors express the same idea when they say that there are no free lunches on Wall Street. It is very difficult to detect undervalued stocks and to beat the market consistently. Of course, this does not mean that investors have perfect foresight and never make mistakes. The large fluctuations in stock market prices suggest that investors are being caught by surprise every day, and that with the benefit of hindsight (but only with that benefit) wonderful profit opportunities were dangling from every tree.

If a stock is fairly priced, it will reflect all existing information and respond only to new information. But new information is necessarily unanticipated. Therefore, the resulting change in the stock price must be a surprise and cannot be predicted from earlier changes in the stock price. So in a market where stocks are fairly valued, their prices will wander randomly. Stocks will be equally likely to offer a high or low return on any particular day, regardless of what occurred on previous days. In other words, prices will follow a random walk.

If you are not sure what we mean by "random walk," consider the following example: You are given \$100 to play a game. At the end of each week, a coin is tossed. If it comes up heads, you win 3% of your investment; if it is tails, you lose 2.5%. Therefore, your payoff at the end of the first week is either \$103 or \$97.50. At the end of the second week, the coin is tossed again. Now the possible outcomes are as follows:



This process is a random walk because successive changes in the value of your stake are independent and determined by the flip of a fair coin. That is, the odds of making money each week are the same, regardless of the value at the start of the week or the pattern of heads or tails in the previous weeks. If a stock's price follows a random walk, the odds of an increase or decrease during any day, month, or year do not depend at all on the stock's previous price moves. In this case, the patterns that investors sometimes believe they can see in stock prices are a mirage, and the historical path of prices gives no useful information about the future—just as a long series of recorded heads and tails gives no information about the next toss.

The notion that successive stock prices follow something close to a random walk has been confirmed in countless ways. Look, for example, at Figure 7.7a. The horizontal axis shows the return on the New York Composite Index in one week (five business days), while the vertical axis shows the return in the following week. Each point in the chart represents a different week over a 40-year period. If a market rise one week tended to be followed by a rise the next week, the points in the chart would plot along an upward-sloping line. But you can see that there was no such tendency; the points are scattered randomly across the chart. Statisticians sometimes measure the relationship between these changes by the correlation coefficient. In our example, the correlation between the market movements in successive weeks is -0.022 . In other words, effectively zero. Figure 7.7b shows a similar plot for monthly (20-business-day) moves. Again, you can see that this month's change in the index gives you almost no clue as to the likely change next month. The correlation between successive monthly changes is -0.004 . This is almost exactly what we would expect in a market where new information is rapidly impounded in stock prices.

Does it surprise you that stocks seem to follow a random walk? If so, imagine that it was not the case and that changes in stock prices were expected to persist for several months. Figure 7.8 provides a hypothetical example of such a predictable cycle. You can see that an upswing in the market started when the index was 3,200 and is expected to carry the price to 3,600 next month. What will happen when investors perceive this bonanza? Because stocks are a bargain at their current level, investors will rush to buy and, in so doing, will push up prices. They will stop buying only when stocks are fairly priced. Thus, as soon as a cycle becomes apparent to investors, they immediately eliminate it by their trading.

Economists often refer to the stock market as an **efficient market**. By this they mean that the competition to find misvalued stocks is intense. So when new information comes out, investors rush to take advantage of it and thereby eliminate any profit opportunities. In other words, the market processes all available information efficiently. As a result, each new change in the price of a stock is independent of earlier changes.

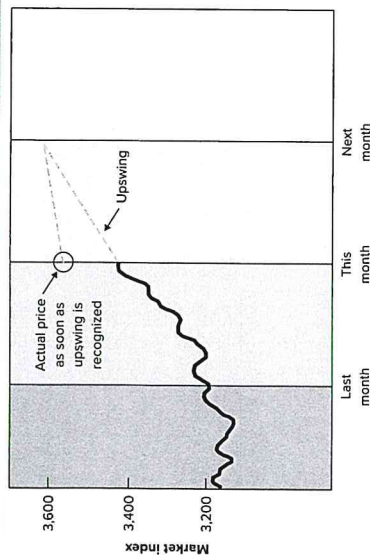
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Stock prices can appear to have patterns
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efficient market
Market in which prices reflect all available information.

FIGURE 7.8 Cycles self-destruct as soon as they are recognized by investors. The stock price instantaneously jumps to the present value of the expected future price.



7.8 Self-Test

True or false: If stock prices follow a random walk,

- successive stock prices are not related.
- successive stock price changes are not related.
- stock prices fluctuate above and below a normal long-run price.
- the history of stock prices cannot be used to predict future returns to investors.

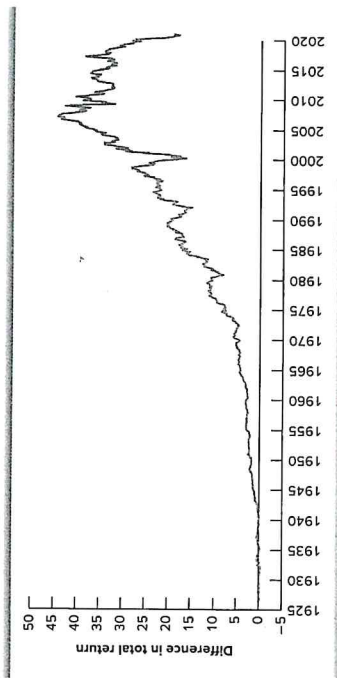
7.7

Market Anomalies and Behavioral Finance

Market Anomalies

Almost without exception, early researchers concluded that the efficient-market hypothesis was a remarkably good description of reality. They tested three versions of the hypothesis. The *weak form* of the hypothesis maintains that stock prices follow a random walk, so that it is impossible to make superior returns just from a study of past price changes. Figure 7.7, which shows that price changes are essentially independent from one period to the next, provides just one example of a test of the weak-form hypothesis. The *semi-strong form* of the hypothesis maintains that public information is impounded in the stock price, so that it is impossible to earn superior returns by a study of information that is also available to other investors. Researchers have tested this hypothesis by looking at how rapidly prices incorporate different types of news such as our example of the CNBC Morning and Midday Calls. Similarly, investment strategies using published information such as company financial statements should be futile. Figure 7.6, which shows that even professional managers (with the greatest access to a wide range of public information) do not generally outperform the market, offers some evidence that the market is semi-strong-form efficient. Finally, the *strong form* of the hypothesis holds that no group of investors, even corporate insiders such as executives or board members, can earn consistently superior returns. This is an extreme version of the hypothesis,

FIGURE 7.9 Difference in the cumulative values of \$1 investments in stocks with high versus low ratios of book value to market value. 1926 = \$0.



since, if true, it implies that even corporate insiders can't exploit their privileged access to information. No one really believes that this version describes the real world perfectly—in fact, legal restrictions against insider trading are designed to prevent insiders from exploiting their obvious informational advantages at the expense of other traders.

However, just as support for the efficient market hypothesis seemed to be airtight, cracks in the evidence began to appear, and soon the finance journals were packed with examples of anomalies, or seeming profit opportunities, that investors have apparently failed to exploit. Some of these concern very short-term stock returns. To have any chance of making money from anomalies that may last for only a few seconds, you need to be a high-frequency trader with one eye on the computer screen and the other on your annual bonus. If you are a corporate financial manager, these patterns may be intriguing conundrums, but they are unlikely to change major financial decisions about which projects to invest in and how they should be financed. Corporate managers should be more concerned about possible mispricing that lasts months or years.

Economists have unearthed a number of these longer-term anomalies. Here are a couple examples.

The Momentum Factor We said earlier that stock prices follow something close to a random walk. That is true, but there do appear to be some exceptions. For example, researchers who have looked at stock market returns over long periods have found a tendency for price rises to persist for some 6 to 9 months and then to revert.¹⁴ Investors refer to this persistence of returns as *momentum*.

The Book-to-Market Factor Now look at Figure 7.9, which shows the cumulative difference between the returns on value stocks and growth stocks. Value stocks are defined as those with a high ratio of book to market value. Growth stocks are those with low ratios of book to market.¹⁵ Notice that value stocks have provided a higher return than growth stocks. Despite the poor relative performance of value stocks since 2017, they still outperformed growth stocks by almost 3.2% annually in the full 1926–2020 period.

Is this pattern another exception to the efficient-market theory, an exception that gave investors the opportunity to earn exceptional returns over many decades simply by investing in low book-to-market stocks? Perhaps. But it also could be that value stocks have extra risks that we have not yet identified or learned how to measure. If so, investors may have demanded the higher returns as compensation for these risks.

¹⁴ See, for example, N. Jegadeesh, "Evidence of Predictable Behavior of Stock Returns," *Journal of Finance* 45, (1990), pp. 881–898; and T. J. Moskowitz, Y. H. Ooi, and L. H. Pedersen, "Time Series Momentum," *Journal of Financial Economics* 104, no. 2 (2012), pp. 228–250.

¹⁵ They are called growth stocks because a high expected rate of growth would justify their high price relative to book value.

Market anomalies such as the momentum factor suggest that there may be occasions when prices of individual stocks can get out of line. But are there also cases in which prices as a whole can no longer be justified by fundamentals? We will look at the evidence in a moment, but first we should note how difficult it is to value common stocks and to determine whether their prices are irrational.

For example, imagine that in December 2019 you wanted to check whether the stocks forming Standard & Poor's Composite Index were fairly valued. As a first stab, you might use the constant-growth formula that we introduced earlier. In 2019, before the Covid pandemic struck in 2020, the annual dividends paid by the companies in the index totaled around \$480 billion. Suppose that these dividends were expected to grow at a steady rate of 4.0% and that investors required a return of 6.0%. Then the constant-growth formula gives a value for the common stocks of

$$PV = \frac{DIV_1}{r - g} = \frac{\$480 \text{ billion}}{.06 - .04} = \$24,000 \text{ billion}$$

which was only slightly less than their actual value of \$26,000 billion in December 2019. But how confident could you be about these figures? Perhaps the likely dividend growth was only 3.5% per year. In that case, your estimate of the value of the common stocks would decline to

$$PV = \frac{DIV_1}{r - g} = \frac{\$480 \text{ billion}}{.06 - .035} = \$19,200 \text{ billion}$$

In other words, a reduction of just half a percentage point in the expected rate of dividend growth would reduce the value of common stocks by 20%. The extreme sensitivity of price to even minor changes in assumptions tells us that valuing common stocks from scratch is always going to be very imprecise. This conclusion has two important consequences.

First, investors find it easier to price a common stock relative to yesterday's price or relative to today's price of comparable securities. In other words, they generally take yesterday's price as correct, adjusting it upward or downward on the basis of today's information. If information arrives smoothly, then, as time passes, investors become increasingly confident that today's price level is correct. But when investors lose confidence in the benchmark of yesterday's price, there may be a period of confused trading and volatile prices before a new benchmark is established.

Second, most of the tests of market efficiency are concerned with relative prices and focus on whether there are easy profits to be made. It is almost impossible to test whether stocks are correctly valued because no one can measure true value with any precision.

Bubbles and Market Efficiency

It may be impossible to prove that market levels are, or are not, consistent with fundamentals. However, every now and again investors seem to be caught up in a speculative frenzy, and asset prices then reach levels that (at least with hindsight) cannot easily be justified by the outlook for profits and dividends. Investors refer to such occasions as bubbles.

The Japanese bubble is a good example. In only four years from 1985 to 1989, the Japanese Nikkei stock index roughly quadrupled. The stock price boom was matched by an even greater explosion in land prices. For example, Ziemba and Schwartz document that the few hundred acres of land under the Emperor's Palace in Tokyo, evaluated at neighborhood land prices, was worth as much as all the land in Canada or California.¹⁶ But then both bubbles burst. By October 1990, the Nikkei had fallen about 50% from its peak, and by March 2009 it was down 80%. Similarly, by 2005 land prices in the six major Japanese cities had slumped to just 13% of their peak.

The dot-com bubble in the United States was almost as dramatic. The technology-heavy NASDAQ stock index rose 580% from the start of 1995 to its eventual high in March 2000. But then, as rapidly as it began, the boom ended, and by October 2002 the index had fallen 78% from its peak.

¹⁶ See W. T. Ziemba and S. L. Schwartz, *Invest Japan* (Chicago: Probus Publishing, 1992), p. 109.

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Two mysterious crashes

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The gyrations in the share price of GameStop in early 2021 were even wilder. Selling at about \$17 per share at the start of the year, the stock garnered both gains and attention when Ryan Cohen, the billionaire founder of Chewy Inc., took a big position in the firm and offered plans for the company to expand its online presence. An army of retail investors who were following the company on the Reddit WallStreetBets site entered into a speculative frenzy, and in a span of just over two weeks, the stock price increased by a multiple of more than 20 to a peak value of \$483. Needless to say, it was hard to justify this run-up based on firm fundamentals.

But when exchanges became increasingly concerned about the high volatility of GameStop and the prospect of broken trades and defaults by investors who had used borrowed money to pay for a portion of their purchases, they tightened both collateral and trading rules, and the bubble burst. Only one week after the stock price peaked, it had fallen to just 11% of that value.

Looking back at these episodes, it seems difficult to believe that expected future cash flows could ever have been sufficient to justify such price run-ups. If that is the case, they present important exceptions to the theory of efficient markets.

But beware of jumping to the conclusion that prices are always arbitrary and capricious. First, most bubbles become obvious only after they have burst. At the time, there often seems to be a plausible explanation for the price run-up. In the dot-com boom, for example, many contemporary observers rationalized stock price gains as justified by the prospect of a new and more profitable economy, driven by technological advances.

Behavioral Finance

Why *might* prices depart from fundamental values? Some scholars believe that the answer to this question lies in behavioral psychology. People are not 100% rational 100% of the time. This shows up in three broad areas—their attitudes to risk, the way that they assess probabilities, and their sentiment about the economy:

1. *Attitudes toward risk.* Psychologists have observed that, when making risky decisions, people are particularly loath to incur losses, even if those losses are small. Losers are liable to regret their actions and kick themselves for having been so foolish. To avoid this unpleasant possibility, individuals will tend to shun those actions that may result in loss.

The pain of a loss seems to depend on whether it comes on the heels of earlier losses. Once investors have suffered a loss, they may be even more cautious not to risk a further loss. Conversely, just as gamblers are known to be more willing to take large bets when they are ahead, so investors may be more prepared to run the risk of a stock market dip after they have experienced a period of substantial gains. If they do then suffer a small loss, they at least have the consolation of being up on the year.

You can see how this sort of behavior could lead to a stock price bubble. For example, early investors in the technology firms that boomed during the dot-com bubble were big winners. They may have stopped worrying about the risk of loss. They may have thrown caution to the winds and piled even more investment into these companies, driving stock prices far above fundamental values. The day of reckoning came when investors woke up and realized how far above fundamental value prices had soared.

2. *Beliefs about probabilities.* Most investors do not have a Ph.D. in probability theory and may make common errors in assessing the probability of uncertain outcomes. Psychologists have found that, when judging the possible future outcomes, individuals commonly look back to what has happened in recent periods and then assume that this is representative of what may occur in the future. The temptation is to project recent experience into the future and to forget the lessons learned from the more distant past. For example, an investor who places too much weight on recent events may judge that glamorous growth companies are very likely to continue to grow rapidly, even though very high rates of growth cannot persist indefinitely.

A second common bias is overconfidence. Most of us believe that we are better-than-average drivers, and most investors think that they are better-than-average stock pickers. We know that two speculators who trade with one another cannot

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Overconfident CFOs

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both make money from the deal; for every winner there must be a loser. But presumably investors are prepared to continue trading because each is confident that it is the other one who is the patsy.

You can see how such characteristics may have reinforced the dot-com boom. As the bull market developed, it generated increased optimism about the future and stimulated demand for shares. The more that investors racked up profits on their stocks, the more confident they became in their views and the more willing they became to bear the risk that the next month might not be so good.

3. *Sentiment.* Efficient markets bring to mind absolutely rational investors on the lookout for every possible profit opportunity. But real investors are people, and people are subject to emotion. Sentiment may be interpreted as their general level of optimism or pessimism about the economy. If the sentiment of large groups of investors varies in tandem, it potentially could have a noticeable impact on security pricing.

Sentiment is offered as another reason that stock market bubbles might develop. An improvement in sentiment can lead to an increase in stock prices. That increase can feed on itself, with investors extrapolating recent performance into the future and deciding that it is a good time to buy, thus pushing prices up even further.

Now it is not difficult to believe that your uncle Harry or aunt Hetty may become caught up in a scatty whirl of irrational exuberance,¹⁷ but why don't hardheaded professional investors bail out of the overpriced stocks and force their prices down to fair value? Perhaps they feel that it is too difficult to predict when the boom will end and that their jobs will be at risk if they move aggressively into cash when others are raking up profits. In this case, sales of stock by the pros are simply not large enough to stem the tide of optimism sweeping the market.

It is too early to say how far behavioral finance scholars can help to sort out some of the puzzles and explain events like the dot-com boom. One thing, however, seems clear: It is relatively easy for statisticians to spot anomalies with the benefit of hindsight and for psychologists to provide an explanation for them. It is much more difficult for investment managers who are at the sharp end to spot and invest in mispriced securities in real time. And that is the basic message of the efficient-market theory. For corporate financial managers there is a clear lesson: Trust market prices unless you have a clear advantage that ensures the odds are in your favor.

SUMMARY

What information is included in stock trading reports, and how can stock-price information about other firms be used to help you value a particular firm? (LO7-1)

- Large companies usually arrange for their stocks to be traded on a stock exchange. The stock listings report the stock's price, price change, trading volume, dividend yield, and **price-earnings (P/E) ratio**.
- To value a stock, financial analysts often start with "valuation by comparables"; that is, they identify similar firms and look at how much investors in these companies are prepared to pay for each dollar of earnings or book assets.

How can one calculate the present value of a stock given forecasts of future dividends and future stock price, and how do a company's growth opportunities show up in its stock price and price-earnings ratio? (LO7-2)

- Stockholders generally expect to receive (1) cash dividends and (2) capital gains or losses. The rate of return that they expect over the next year is the expected dividend per share DIV_1 plus the expected increase in price $P_1 - P_0$, all divided by the price at the start of the year P_0 .

¹⁷ The term "irrational exuberance" was coined by Alan Greenspan, former chairman of the Federal Reserve Board, to describe the dot-com boom. It was also the title of a book by Robert Shiller that examined the boom. See R. Shiller, *Irrational Exuberance* (New York City: Broadway Books, 2001).

- The present value of a share is equal to the stream of expected dividends per share up to some horizon date plus the expected price at this date, all discounted at investors' required rate of return. Required return is the return one can expect to earn on other stocks posing the same degree of risk.
- If the horizon date is far away, the present value of the final stock price will be small, so we simply say that stock price equals the present value of all future dividends per share. This is the **dividend discount model**.

- If forecast dividends grow at a constant rate, g , then the value of a stock is $P_0 = DIV_1 / (r - g)$. This formula is called the **constant-growth dividend discount model**.
- When the company is currently not in a stable growth phase, but may eventually enter one, we may discount forecast dividends for the next few years and then discount the expected price at some horizon date when the growth rate is expected to stabilize. This expected price is often estimated using the constant-growth model, applied at that horizon date.
- You can think of a share's value as the sum of two parts—the value of the assets in place and the **present value of growth opportunities**—that is, of future opportunities for the firm to invest in high-return projects. The price-earnings (P/E) ratio reflects the market's assessment of the firm's growth opportunities.

How can stock valuation formulas be used to infer the value of an entire business? (LO7-3)

- When we try to estimate the value of an entire business, instead of discounting projected dividends, we use the free cash flows provided by the business. Free cash flows are those that are not plowed back into the business and therefore would be available to pay dividends or repurchase stock.

How does competition between investors lead to efficient markets? (LO7-4)

- Competition between investors will tend to produce an **efficient market**—that is, a market in which prices rapidly reflect new information and investors have difficulty making consistently superior returns. If the market is efficient, all investors can rationally expect a return that is sufficient (on average) to compensate for the time value of money as well as investment risk.
- The evidence for market efficiency is voluminous, and there is little doubt that skilled professional investors find it difficult to beat market indexes consistently. Nevertheless, there remain some puzzling instances where markets do not seem to be efficient. Some financial economists attribute these apparent anomalies to behavioral foibles.

LISTING OF EQUATIONS

$$7.1 \quad V_0 = \frac{DIV_1 + P_1}{1 + r}$$

$$7.2 \quad P_0 = \frac{DIV_1}{1 + r} + \frac{DIV_2}{(1 + r)^2} + \dots + \frac{DIV_H + P_H}{(1 + r)^H}$$

$$7.3 \quad P_0 = \frac{DIV_1}{r - g} \quad (\text{Constant-growth dividend discount model})$$

$$7.4 \quad r = \frac{DIV_1}{P_0} + g = \text{dividend yield} + \text{growth rate}$$

$$7.5 \quad \text{Sustainable growth rate} = \text{return on equity} \times \text{plowback ratio}$$

QUESTIONS AND PROBLEMS



1. Stock Markets. True or false? (LO7-1)

- a. The bid price is always greater than the ask price.
- b. An investor who wants to sell his stock immediately should enter a limit order.
- c. The sale of shares by a large investor is usually termed a "primary offering."
- d. Electronic Communications Network refers to the automated ticker tape on the New York Stock Exchange.